Mefa

Important Formulas

Unit 1

(1) GDP = within a country border, during one year ~~final~~ total value (cost) of final goods / services.

(2) NDP = GDP − Depreciation.

(3) GNP = GDP + Net factor Income from Abroad

(4) NNP = GNP − Depreciation

(5) PI → Salary, rent, interest etc.

(6) Private income → earned by private individual & enterprices within contry

(7) DPI = PI − Direct tax

(8) PCI = $\frac{NI}{\text{Total Population}}$

Unit 2

Demand function =

$$Q_d = f(P, Y, P_T, T, E) = \boxed{Q = a - bp}$$

Here Q_d = Quantity demand

P = Price of product

Y = Income of Consumer

P_T = Price of goods

T = Taste / Preference

E = Expectations

$Q_d = \infty \rightarrow$ Perfect elastic

$Q_d => 1 \Rightarrow$ elastic

$Q_d = 1 \Rightarrow$ Unitary

$Q_d < 1 \rightarrow$ Inelastic

$Q_d = 0$ Perfect inelastic

(1) Price Elasticity of demand (E_p) = $\dfrac{\text{\% change in quantity demand}}{\text{\% chang in price}}$

$$E_p = \frac{dQ/Q}{dP/P}$$

(2) Income elasticity of demand = $\dfrac{\text{\% change in quantity demand}}{\text{\% change in income}}$

$$(E_y) = \frac{dQ/Q}{dI/I}$$

(3) Advertising elasticity of demand = $\dfrac{\text{\% change in quantity demand}}{\text{\% change in advertising expenditure}}$

$$(E_a) = \frac{dQ/Q}{dA/A}$$

(4) Cross elasticity of demand (E_c) = $\dfrac{\text{\% change in quantity demand of Commodity } x}{\text{\% change in price of commodity } y}$

$$E_c = \frac{dQ_x/dQ}{dP_y/d.}$$

$$\boxed{E_c = \frac{dQ_x/Q_x}{dP_y/P_y}}$$

Relation between all elasticity.

$$\boxed{E_P = KX \cdot E_Y + (1 - KX) E_T}$$

E_P = Price elasticity
E_Y = Income elasticity
E_T = Substitute
KX = Consumer Income.
~~1-KX~~ =

Elasticity of Supply

$$E_s = \frac{\text{\% change in Quantity supply}}{\text{\% change in Price}}$$

$$\boxed{E_s = \frac{dQ_s / Q_s}{dP / P}}$$

Supply Function

$$Q_s = f(P_n, P_r, C, T, N, G, E)$$

where Q_s = Quantity supplied
P_n = Price of good
P_r = related Price of good
C = Cost of Production
T = Technology
N = Number of sellers
E = Expectations.

$E_s = \infty \rightarrow$ Perfectly elastic supply
$E_s > 1 \rightarrow$ Elastic supply
$E_s = 1 \Rightarrow$ Unitary supply.
$E_s < 1 \Rightarrow$ Inelastic supply.

($E_s = 0$ Perfect inelastic)

Production

Production function

$$f_Q = f(L, K, N, T)$$

Here Q = output

L = Labour , N = land / Natural

K = Capital , T = Technology expenditure.

For long term - all factors are variable

$$\boxed{Q = AL^a K^b N^c T^d}$$

for short run - atleast one is fixed (Capital)

$$\boxed{Q = AL^a N^c T^d}$$

$$\boxed{\text{Cobb-douglas} \rightarrow Q = AL^\alpha K^\beta}$$

Basic Concept

long run

Stage

(1) Icreasing Return

output ↑ more the input

(2) Constant Return

output increase same the input.

(3) Decreasing Return

output increase Less then inputs

Short Run

Stage

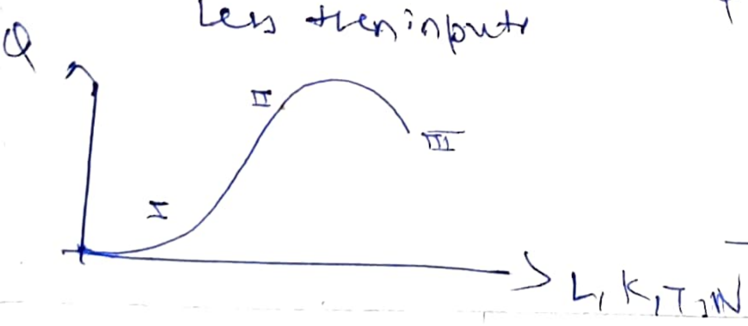
Increasing return

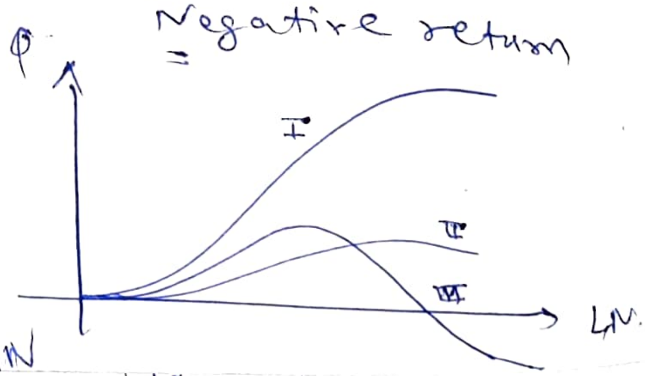
Stage 2

Diminishing return (most efficient stage)

Stage 3

Negative return





** Cost

* Implicit => economic cost = which causes of profit

* Explicit => accounting cost = actual cost which we make payment of product.

* opportunity → next best alternative that is also part of implicit cost.
  Imp decision making.

* Private ~~cost~~ → Personal income.
  Social cost => Paid by government

* Shut down cost = Cost in temporary session. (rental cost)
  abandonment cost = Cost of retiring a fixed asset from use.
  (ex. selling a land)

* Sunk cost = Cost that already incurred & cannot be recovered.
  ~~डूबी हुई~~ (डूबी हुई Cost)

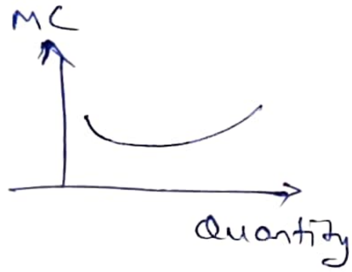
* direct → ~~impli~~ explicit.
  Indirect → implicit

Cost function in Short Run

* TFC → Fixed cost like market prices
* TVC = varies with factor of production (like labour, capital)
* TC = TFC + TVC
* Average fixed cost = $\dfrac{TFC}{\text{level of output}}$
* Average variable cost = $\dfrac{TVC}{\text{Quantity of Product}}$
* Marginal Cost = Additional cost.

$$MC = \frac{\text{Total cost} -}{\text{output}}$$

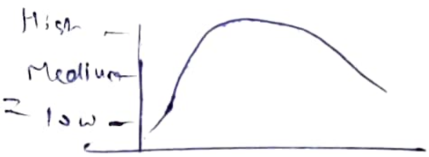
MC curve



Cost function in Long Run

* Average Cost
* Marginal Cost ⇒ $TC_m - TC_{m-1}$

Imp Isoquants Curve =



Combination of all possible input (labour, capital) that produce same level output

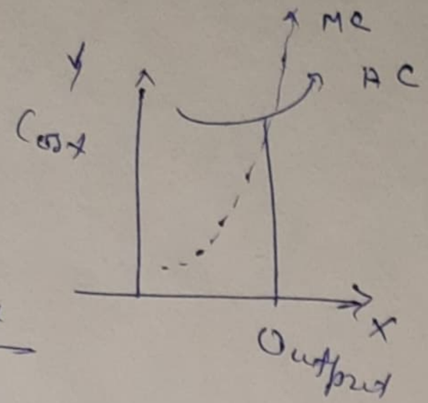
Marginal Rate of Techinacal substitution.

$$MRTS = \frac{MP_L}{MP_K} = \left(\frac{\alpha}{1-\alpha}\right) \cdot K/L$$

K = Kapital
L = labour

MP = Marginal Product.

Variable Proportion



(i) Total Product = total quantity of output produced.

(ii) Marginal Product = $\frac{\text{Total Product}}{\text{Labour}}$

(iii) Average Product (AP) = $\frac{\text{Total Product}}{\text{Units of variable factor}}$

~~Unit 4 Market Structure~~

* Least cost combination input.

$$\frac{MP_L}{P_L} = \frac{MP_K}{P_K}$$